Fiscal Year:	FY 2010	**Task Last Updated:**	FY 02/03/2011
PI Name:	Fitts, Robert H Ph.D.		
Project Title:	Effect of Prolonged Space Flight on Human Skeletal Muscle		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53201	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	10/01/1998	**End Date:**	09/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS Increments 5, 6, 7, 9, 10, and 11 (per PI 6/07) Data collection complete, analysis in progress (6/2007) NOTE: Received extension to 9/30/2010 per PI (4/2010) NOTE: Received NCE to 03/30/2010 per J. Dardano/JSC (9/2009) NOTE: Received NCE to 9/30/2009 per PI (1/2009) NOTE: Received NCE to 09/30/2008, per PI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Trappe, Scott (Ball State University) Riley, Danny (Medical College of Wisconsin)		
Grant/Contract No.:	NCC9-116		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our long term goals are to understand the effects of zero g space flight on the neuromuscular system, and work to develop effective exercise countermeasures. The specific goals of this project are to establish a time course for the structural and functional alterations in limb skeletal muscle with prolonged weightlessness, and from these data calculate the new microgravity steady state that would likely be reached in a trip to Mars. Since we have already made a detailed characterization of how human limb muscles react to short duration (17 days) flight, we propose to study the effects of 3 flights of progressively longer duration. The ideal design would be to study 30, 90, 180 or 30, 60, 90 days. The ideal number of subjects for each flight would be 3 to 4, but we realize from experience that the number available for a given duration flight maybe 1 or 2. This project is designed as a pre- and post-flight study. Calf muscle function will be evaluated using a specially designed torque velocity dynamometer, and biopsies will be taken from the soleus and gastrocnemius muscles at L-45 and immediately post flight (R+0). The effect of prolonged space flight on the functional and structural properties of individual fast and slow fibers will be determined with special emphasis on fiber type specific changes in cell size, peak force and power, the maximal shortening velocity, and the cells resistance to damage. Immunohistochemical studies will be carried out to assess the extent of slow to fast isozyme shifts in myosin, and EM analyses will determine the structural relationship of the thick and thin filament, the degree of myofilament loss, and the integrity of the dystrophin-dystroglycan complex. Microbiochemical studies will establish the extent to which prolonged zero-g affects the substrate and enzyme profile of each fiber type. In conclusion, we will establish the time course for functional and structural change in limb skeletal muscle with prolonged space flight, and establish mechanistic explanations for many of the observed changes. At a minimum we will establish the relative importance of changes in the myosin and actin filament structure vs qualitative changes in the myosin protein, and the relationship of the integrity of the dystrophin-dystroglycan complex to the ability of the cell to resist stretch-induced damage. See also http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary benefit will be to allow countermeasures to be developed that will prevent the microgravity-induced loss of muscle mass and function, such that, prolonged missions to Mars and beyond can be undertaken. This project will establish the time course of the loss of muscle mass and function with microgravity durations from 30 to 180 days. The results of these studies will have direct benefits to conditions on earth that produce long periods of inactivity, such as, illnesses requiring bed rest, and falls resulting in limb immobilization- the latter is a particularly prevalent problem in the elderly. Of particular importance will be the studies designed to establish the cause of muscle tearing as damage of this sort is a common occurrence in the elderly. Additionally, the results will provide the cellular and molecular data needed to develop effective exercises for slowing the loss of muscle mass that occur with aging.
Task Progress:	The analysis of data from ISS Increments 5-11 has been completed and the data set in excel files was sent to NASA for archiving. Two peer-reviewed papers have been published and a third is in preparation. The paper in the Journal of Physiology volume 588, pp 3567-3592, 2010 entitled "Prolonged space flight-induced alterations in the structure and function of human skeletal muscle fibres" was reported in over 200 newspapers world wide and the PI was interviewed on the BBC, Canadian, and New Zealand public radio and Japan National TV. Our 2009 peer-reviewed paper in the Journal of Applied Physiology vol 106, pp. 1159-1168 was featured in the Journal of Aviation, Space, and Environmental Medicine vol 81, N0 1, pp. 1050-1052 in the section entitled Research Progress Reports from NASA Human Research Progress. The main findings and implications of our project are: 1. The exercise program on the ISS did not completely protect the calf muscles. We observed a substantial decrease in calf muscle mass and performance along with a slow-to-fast fiber type transition in the gastrocnemius and soleus muscles, which are all traits associated with unloading in humans. 2. Based on comparisons with long duration bed rest studies without countermeasures, the exercise in space is having a beneficial effect but is not complete, with the soleus being more difficult to protect than the gastrocnemius. 3. Four factors (in addition to the microgravity environment) appear to contribute to the reduced calf muscle volume: a) the initial calf muscle volume as crewmembers with the largest muscles showed the greatest atrophy. This factor is a direct result of the inadequacy of the high resistance exercise countermeasure such that resistance trained individuals with large muscles are unable to maintain them; b) the volume of treadmill exercise as running protects the calf muscles better than cycle ergometry; c) the volume of treadmill running in the passive mode as more force is needed to drive the belt ; and d) inadequate caloric intake which likely contributed to the muscle atrophy. A previous energy expenditure report of humans working in space with no exercise countermeasures was ~ 31 kcal•kg-1•day-1, while the average daily energy intake of the crewmembers in this study was ~ 30 kcal•kg-1•day-1. This is 17-23% below the predicted caloric need of 35-37 kcal•kg-1•day-1 when the exercise is considered. 4. This is the first study providing a cellular analysis of the effects of long duration space flight on the structure and function of human skeletal muscle. Importantly, prolonged weightlessness produced substantial loss of fiber mass, force and power with the hierarchy of the effects being soleus type I > soleus type II > gastrocnemius type I > gastrocnemius type II. The post-flight decline in power was attributed to declines in both force and velocity. 5. Structurally, the quantitatively most important adaptation was fiber atrophy, and the resulting loss and shrinkage of myofibrils. An increased packing density of the remaining thick and thin filaments is likely to have contributed to the reduced fiber velocity, and increased fiber stiffness. 6. The decline in calf muscle volume, reduced calf force and power, and the altered structure and function of the individual fibers document the inadequacy of the current exercise countermeasures, and point to the need for the development of better exercise protocols and hardware for preventing undesirable transformation of fiber structure and function. 7. The aerobic exercise (treadmill and cycle ergometry) protected the muscle aerobic enzyme capacity, but did not prevent fiber atrophy and loss of force, velocity and power. Based on conditioning biographies (Trappe et al. 2009) and type I fiber aerobic enzyme capacity, the crewmembers in the “low treadmill” group were in better aerobic condition at the start of the mission compared to the “high treadmill” group. Thus, the greater decline in fiber size and force of the

	"low treadmill" group likely reflects the combined effects of greater initial fitness (both aerobic and strength), and the ineffectiveness of the in-flight exercise countermeasures. 8. Presumably, with a better in-flight high resistance training device, crewmembers would be able to maintain muscle mass and strength gained through pre-flight exercise programs. 9. Clearly, the deleterious changes in muscle structure and function with prolonged space flight documented in this study have potentially serious implications for a variety of circumstances including but not limited to the crew's ability to perform an emergency egress; muscle damage due to high force particularly eccentric contractions; and difficulties performing work on the surface of the moon or Mars. 10. Developing new high resistance exercise countermeasures, documenting their effectiveness on the ISS, and monitoring crew member nutritional status to insure an adequate caloric intake should be the high priorities that fulfills the intended scientific use of the spacecraft. Publications. 1. Trappe, S., D. Costill, P. Gallagher, A. Creer, J.R. Peters, H. Evans, D.A. Riley, and R.H. Fitts. 2009. Exercise in space: human skeletal muscle after 6 months aboard the International Space Station. J. Appl. Physiol. 106: 1159-1168. 2. Fitts, R.H., P.A. Colloton, J.G. Romatowski, J.R. Peters, S.W. Trappe, D.L. Costill, J.L. Bain, and D.A. Riley. 2010. Prolonged Space Flight-Induced Alterations in the Structure and Function of Human Skeletal Muscle Fibres. J. Physiol. (Lond). 588: 3567-3592.
Bibliography Type:	Description: (Last Updated: 01/28/2011)
Articles in Peer-reviewed Journals	Fitts RH, Trappe SW, Costill DL, Gallagher PM, Creer AC, Colloton PA, Peters JR, Romatowski JG, Bain JL, Riley DA. "Prolonged space flight-induced alterations in the structure and function of human skeletal muscle fibres." J Physiol. 2010 Sep 15;588(Pt 18):3567-92. Epub 2010 Jul 26. PMID: 20660569 , Sep-2010
Articles in Peer-reviewed Journals	Trappe S, Costill D, Gallagher P, Creer A, Peters JR, Evans H, Riley DA, Fitts RH. "Exercise in space: human skeletal muscle after 6 months aboard the International Space Station." J Appl Physiol. 2009 Apr;106(4):1159-68. Epub 2009 Jan 15. PMID: 19150852 , Apr-2009